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Imbalanced Data Classification Modelling Using CTGAN and Decision Tree for Student Graduation Predicting in a Courses

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***Abstract*: -** Student graduation in one course is the main factor in supporting the learning process and minimizing the occurrence of drop outs. In this case, a prediction model is needed to be able to identify student graduation at the beginning of the learning process. The aim of this research is to produce a prediction model that has significant accuracy in predicting student graduation in one course using the decision tree algorithm and implementing the conditional tabular generative adversarial networks (CTGAN) model. CTGAN is a model that can produce synthetic data on certain input variables. First, the graduation dataset is collected and pre-processed, then a labeling process is carried out on the dataset, so that the data can be used as initial input for CTGAN modeling. Next, the dataset with certain label features is subjected to an oversampling process using the CTGAN model. Finally, a prediction process was carried out using a decision tree algorithm to produce significant accuracy values by utilizing the results of data processing with the CTGAN model. The results show that the decision tree algorithm has a very significant accuracy value in predicting student graduation performance using the CTGAN model processing dataset, as well as a significant precision value in predicting student failure in the learning process of a course.

***Keywords*:** Student Graduation, Decision Tree, Data Oversampling, CTGAN, Accuracy, Precision, Recall, f1-skor.

1. Introduction

The problem of dropout in higher education is very prominent. When dropout increases, higher education will lose the number of students, affecting academic performance and the implementation of education in higher education. Carrying out an initial identification process for students who are at risk of failing in the learning process will prevent an increase in the dropout rate[1]. The process of predicting student learning performance is the main stage which can not only help students control independent learning, but can identify students who are at risk and reduce students who fail in the learning process[2]. The use of predictive modelling methods can help to identify students who are at risk in the learning process from the start[3]. Predicting student academic performance is the main factor for educators to get an initial response and take various solutions as preventive measures to improve student academic performance[4]. Academic achievement is the learning outcomes achieved by students and is the most important factor in representing the quality of academic performance in completing the learning process. Academic performance is one of the main factors for evaluating the learning process, teaching process, and improving learning materials so that students can increase their minimum learning competency achievements in a course[5]. The assessment feature of academic performance can include the main demographics of students, the scores on each supporting exam component, and the final score of the learning process can provide information about the results of academic performance achievements which can be used as a basis for predicting graduation in a course[6]. Carrying out a prediction process for academic performance can improve students' academic achievements by providing solutions based on the predicted graduation results produced in the learning process. According to[7], the main objective in the prediction process for student graduation is to increase learning success and learning motivation to complete learning on time. The ability of

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higher education to predict the success of student academic achievement in the learning system is very important process, the aim is to improve the quality and competence of students and prevent an increase in the dropout rate[8]. Predicting a student's academic performance which is carried out at the beginning is very important for student success achievements in the learning process so that it can easily identify student failures in carrying out the learning process and can also provide appropriate solutions[9]. One of the obstacles in higher education institutions is the ability to provide quality education to students, so educational institutions must be able to develop various innovative methodologies to help improve the quality of student studies and facilitate student graduation in the learning process[10]. Machine Learning is a part of Artificial Intelligence (AI), which can process various data to generalize data, predict results, and measure the performance of models and effective computing systems to produce accurate decisions in developments in the education sector. One of the uses of machine learning algorithms in the education sector is to solve the problem of imbalanced data graduation student data in higher education. Most of the problems in processing data on student learning outcomes in predicting passing a course are unbalanced, thus affecting the performance of the prediction model, problems with data imbalance occur when the distribution of classes is uneven in a collection of datasets[11]. Implementation of data processing with data mining and machine learning can be used in the education sector to produce significant predictive performance through the identification, extraction, and evaluation of factors related to the student learning process[12]. Predicting student success in the learning process by implementing machine learning models, aims to develop models and explore information to produce more effective learning performance to improve students' academic rankings and achievements[13]. Data processing using a data mining approach in the education sector has provided positive results in developing a prediction model for student success in the learning process based on performance characteristics, and learning preferences to achieve the expected results[14]. Machine learning algorithms can be used to predict students who are at risk and likely to drop out of school, by utilizing log data such as quizzes, attendance, exams, and grades, to evaluate and predict academic performance in the learning process[15]. One of the machine learning methods used to make predictions is a decision tree. The decision tree algorithm is used to assess the accuracy of the performance of prediction models based on learning preferences, academic learning achievements so that the resulting decisions have an impact on increasing student learning competence[16]. The decision tree algorithm can produce classification and predictive models that can be easily understood, visualized, and displayed in graphical form so that it is easy to carry out data analysis[17] The decision tree algorithm is a classification model that is often used, and can also be used to implement data-based predictive models by representing each data at each decision node to express the resulting information[18]. The CTGAN model is one solution to overcome the problem of modelling tabular data in minority classes to balance the dataset and increase the accuracy of the performance of the prediction model on student graduation in one course. The availability of unbalanced data will have an impact on the performance of the resulting model, making data processing more complicated, therefore the conditional tabular generative adversarial networks (CTGAN) model approach is used to handle the problem of unbalanced data in a collection of datasets[19]. Overcoming the problem of imbalanced data with the CTGAN model has had a positive impact in producing synthetic data sets needed for developing a course graduation prediction model[20]. In many cases, the limited experimental data available is one of the main problems in implementing machine learning algorithms to deal with imbalanced data. Therefore, the approach using the conditional tabular generative adversarial networks (CTGAN) model and decision tree algorithm in predicting student graduation in a course is the main discussion in this research.

2. Related Work

The existence of a dataset in the research domain is the main thing that must be prepared. The success of a model depends on the availability of data to be processed. A collection of datasets with a normal distribution will certainly provide better model performance results. In various studies in the field of education, the prediction process has provided benefits in improving academic performance. Still, the data sets used in the prediction process are often found to be imbalanced. To overcome this, one of the models used is CTGAN. CTGAN can separate majority data as main data and minority data as outlier data or new data which is often called data in the non-passing category. Decision Tree is a classification model that can not only be an effective method for producing analyses of student learning performance and effectively processing data[21]. Decision Tree is a form of predictive modeling that is generally used in various fields with an algorithmic approach that can identify data sets based on student preferences as users[22]. The Implementation of the prediction model using the decision tree approach provides significant results with a success rate reaching 91% which can accurately predict the success of student academic achievement in the learning process, and is also very effective in improving the performance of prediction models on imbalanced data sets[23]. One of the advantages of the decision tree algorithm is the ability to interpret the knowledge that is extracted and conveyed in the form of classification rules or visualization of classification data in the form of a tree structure where each rule is a representation of each tree path from the root to each leaf[24]. The decision tree algorithm can produce significant prediction values, where the information shown can be represented in the form of a hierarchical tree, this can be seen in Figure 1[25].

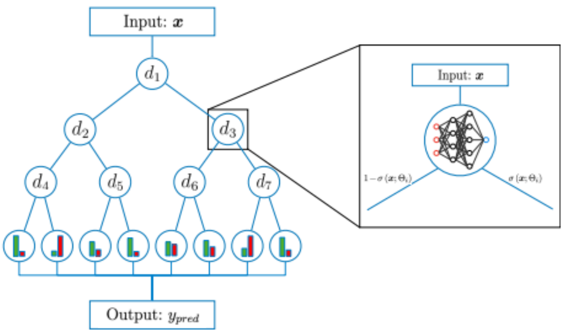

Fig. 1. illustration of decision tree algorithm

Each model in machine learning can have its performance measured to determine the model's ability to carry out data processing, prediction, and classification. One of the models used to make predictions is the decision tree model. To find out the performance of the decision tree model, it can be done by calculating the accuracy, precision and recall values. The results of these components can be used as an analytical tool to state that the model used is appropriate. According to [26], that one method used to measure model performance in predicting student success in one course is to calculate the accuracy, precision, and recall components. Accuracy is calculated by comparing the identification results of students who passed with the total number of students who passed and did not pass. Another explanation can be seen in the equation below:

$$Accuracy - \frac{TP+TN}{TP+TN+FP+FN} \quad \dots\dots\dots\dots\dots\dots (1)$$

Noted: TP = True Positive, TN = True Negative, FP = False Positive, FN = False Negative

$$Precision - \frac{TP}{TP+FP} \dots\dots\dots\dots\dots\dots (2)$$

Noted: TP = True Positive, FP = False Positive

$$Recall - \frac{TP}{TP+FN} \quad \dots\dots\dots\dots\dots\dots (3)$$

Noted: TP = True Positive, FN = False Negative

$$F1\ score - 2 * \frac{Precision * Recall}{TPrecision + Recall} \quad \dots\dots\dots\dots\dots\dots (4)$$

The main problem in predicting student failure in one course is the collection of data on student academic achievements, where the data collection found is unbalanced. One model used to handle unbalanced data is CTGAN. The conditional tabular generative adversarial network (CTGAN) model is divided into two neural network modules, namely the generator (G) and the Critic (C), the generator (G) is used to study conditional distributions aimed at real data to produce synthetic data and the critical (C) is used to separate actual data from synthetic data so that it can produce an amount of data that is close to the actual data[27]. The CTGAN model can improve the balance between data classes with the same size as the original dataset but with a more balanced distribution of labels, so it can provide more data in the training process to help machine learning algorithms decipher patterns and produce data samples that suit data processing needs[28]. In the data processing process using machine learning, most of the data sets are balanced so that the level of pattern recognition and measuring model performance is better. Unbalanced data availability largely affects model performance and makes the data processing process in machine learning algorithms have a low level of identification and validation performance so that the Conditional Tabular Generative Adversarial Network (CTGAN) algorithm is needed to expand and balance the amount of data to be used in the training and prediction process[29]. CTGAN Model can be seen in Fig. 2, is a model that can produce realistic synthetic data by using a Generative Adversarial Network and applying specific mode normalization to overcome the number of non-Gaussian and multimodal distributions so that it can increase the accuracy of the model and the resulting synthetic data can be used to train a learning-based recommendation system[30].

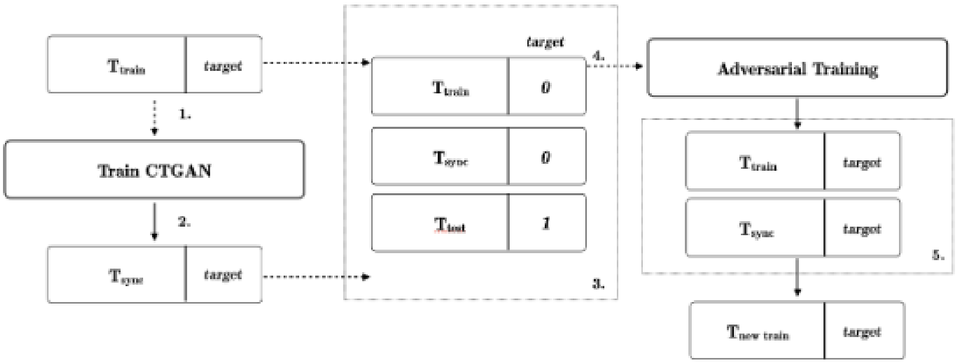

Fig. 2. Design and workflow of CTGAN implementation.

The use of the CTGAN model in various studies that utilize data sets with imbalanced data categories is considered more effective because the CTGAN model uses synthetic minority oversampling techniques to produce synthetic data and can also improve the performance of prediction and pattern recognition models[31]. CTGAN modeling is a type of tabular data modeling to overcome unbalanced data categories by exploring each feature contained in the tabular data structure, where for each feature, the minimum value is represented as 0 or −1 and the maximum value is represented as 1[32].

3. Methodology

3.1 Research Framework

The research framework can be seen in the picture (Fig. 3. Research Framework). In this research, the research framework is divided into several stages, including 1) Input analysis, the stage where the researcher carries out various research activities including literature review, description of the research question, research problem, research purpose, and problem formulation. This stage aims to find solutions to the research problems that will be resolved and data analyse. 2) Data pre-processing, is the stage where researchers prepare a collection of datasets which are the initial input in solving research problems. At this stage, data processing begins with data pre-processing which aims to eliminate several problems in the dataset collection that can interfere with data processing, such as inconsistent data formats, empty data values, and inappropriate data types, 3) Data balancing, At this stage a data balancing process is carried out using the CTGAN model where the aim is to produce data in a collection of datasets, especially data in the non-pass category so that the resulting dataset is balanced. The process of generating unbalanced data into balanced data aims to reduce the time complexity of implementing model performance so that it can improve prediction model performance and produce accurate information. 4) Prediction modelling, this stage explains the nature of the approach taken in solving research problems which aims to analyse data patterns to determine whether new opportunities can be applied. This technique requires a machine learning approach to produce more accurate prediction performance, in this case, the approach used is a decision tree (DT). At this stage, modelling with the DT approach begins with the labelled training dataset and labelled testing dataset stages, then the DT model is trained on the labelled training dataset. 5) Model performance, this stage is the final stage in implementing the DT model, which will produce predicted values for the DT model performance measure in the form of a confusion matrix. The results displayed are in the form of information about accuracy, precision, and recall values which are a representation of the predictions produced. 6) Evaluation, this stage explains the results of the evaluation of the performance of the prediction model using the DT approach, which is measured using a confusion matrix consisting of true positive (TP), false positive (FP), true negative (TN) and false negative (FN) values. which will then be represented in value in the form of precision and accuracy. At this stage, we will also explain the precision value which is the main point in representing prediction results using DT so that researchers can provide accurate information. 7) Reports and publications are the final stages of the research process, where the results obtained will be published in the form of a scientific journal and as a medium for conveying information on research results.

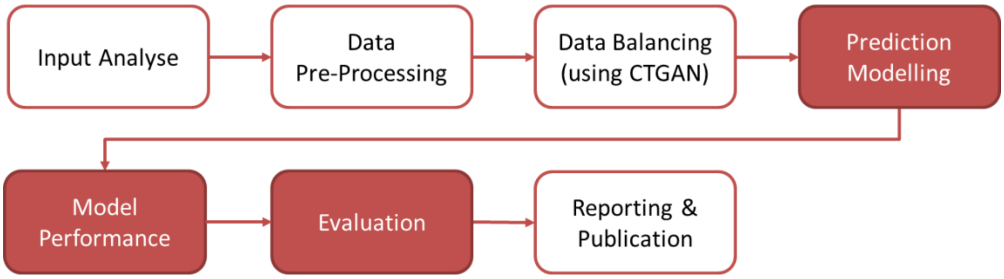

Fig. 3. Research Framework

3.2 Data Balancing with Conditional Tabular Generative Adversarial Network (CTGAN)

In carrying out the process of predicting student graduation in a course, a balanced data set is needed so that it can help the model's performance in producing accurate information. In this research, the data collection on student academic performance is unbalanced data, where data in the pass category is more dominant than data in the fail category. To overcome this, a data balancing process is needed so that predictions of student graduation in one course produce significant accuracy and precision values. One of the models used is the Conditional Tabular Generative Adversarial Network (CTGAN). CTGAN uses GAN-based methods to model the distribution of tabular data and sample rows from that distribution.

In CTGAN, mode-specific normalization techniques are utilized to handle columns containing non-Gaussian and multimodal distributions, while conditional generators and training-by-sampling methods are used to address the class imbalance problem. An illustration of the CTGAN model can be seen in Fig. 4. CTGAN Data Balancing Model.

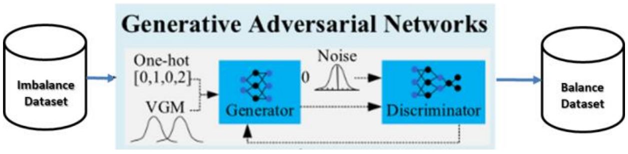

Fig. 4. CTGAN Model Balancing Data

The use of the CTGAN model in predicting student graduation in a course is an imbalance related to the pass-and-fail data category features. In this case, if a data category that does not pass is underrepresented in the dataset, then that category cannot be processed by the generator, so creating synthetic data is one way to falsify the original data and make it into synthetic data to improve the performance of predictive modeling or classification. to produce accurate values. CTGAN introduces Conditional Generators to overcome various problems caused by various unbalanced data categories, especially in handling student graduation data in a course which can cause model performance to be less significant. On this basis, CTGAN considers conditional vectors to use in the dataset.

3.3 Decision Tree Models

In this research, the first thing that can be explained is in carrying out predictions of student graduation in a course using the decision tree approach. In the decision tree model, the tree consists of nodes and branches where each node represents a decision or test result based on specified data features. Meanwhile, branches will connect each node produced in the prediction process. The decision tree model is illustrated in Fig. 5. The Decision Tree Prediction Model for Student Graduation describes the prediction process by producing pass and fail categories based on attendance, quizzes, independent assignments, group assignments, midterms, and final exams. The prediction process using the decision tree approach begins with data modelling where the available dataset is divided into 2 parts, namely training data of 80% and testing data of 20%. At the data modelling stage, the dataset will undergo a labelling process based on data processing needs, after that a data training process will be carried out first to train the data with various patterns in the dataset. And after that, a data testing process will be carried out. Then a prediction process is carried out based on the results of the data modelling to produce information regarding student graduation in a course in learning process.

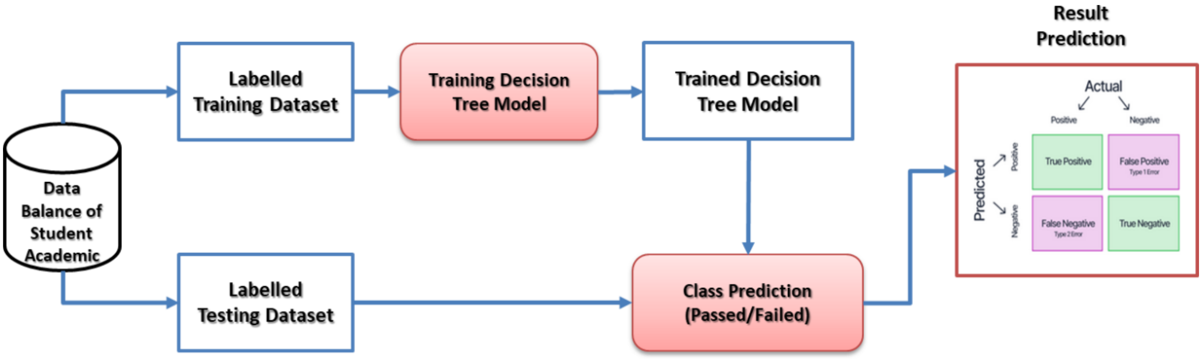

Fig. 5. CTGAN Model Balancing Data

The results achieved in this research are displayed in the form of a confusion matrix to show the performance of the decision tree model on the predicted results of student graduation in a course in the learning process. Apart from that, other results are displayed in the form of a prediction model performance table using a decision tree model consisting of accuracy, precision, recall, and f1-score values which aim to show the results of the model's performance on the data set used in the prediction process using the decision tree approach.

4. Result and Discussion

4.1 Input Analyse

Input analysis is a process to determine the relationship between prediction features consisting of Attendance, group assignments (TK), independent assignments (TM), quizzes (Quis), middle tests, and final tests. The approach taken to determine the relationship between these features used a descriptive analysis model to display the results achieved in the form of a hit map graphic as shown in Fig. 6. Descriptive Analysis of Future Predictor.

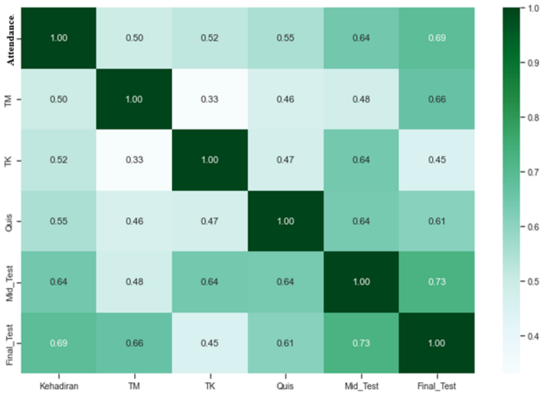

Fig. 6. Descriptive Analysis of Future Predictor

The results shown in the input analysis process show that each feature used as a predictor variable has a relationship that influences each other in the prediction process and determining student graduation in a course in the learning process.

4.2 Data Pre-Processing

Pre-processing is a process in data mining that is used to identify data formats, data types, data availability and data patterns contained in data sets. The main goal of pre-processing is to ensure that the data to be processed is valid and meets the required data format.

```
data.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 738 entries, 0 to 737
Data columns (total 12 columns):
 #   Column              Non-Null Count  Dtype
---  ------              --------------  -----
 0   NIM                 738 non-null    int64
 1   LSI                 738 non-null    object
 2   TPA                 738 non-null    float64
 3   Learning_Interests  738 non-null    float64
 4   Kehadiran           738 non-null    float64
 5   Tugas_Mandiri       738 non-null    float64
 6   Tugas_Kelompok      738 non-null    float64
 7   Quis                738 non-null    float64
 8   Mid_Test            738 non-null    float64
 9   Final_Test          738 non-null    float64
 10  Result              738 non-null    float64
 11  Status              738 non-null    object
dtypes: float64(9), int64(1), object(2)
memory usage: 69.3+ KB
```

Fig. 7. Data Pre-processing Results

The results of pre-processing data in this study show that missing data in the data set was not found and also the format available in the data set was also appropriate and the availability of data in the data set was also significant enough to be used as basic data in the process of predicting student graduation in a course. which is shown in Fig. 7. Data Pre-processing Results.

4.3 Data Balancing Process

The data balancing process is a very important stage in the process of predicting student graduation in a course in the learning process, this is because the availability of data to be processed is unbalanced data. Where the student graduation data set is more dominant in data from the pass category, so it is necessary to carry out a data balancing process. The main objective of the data balancing process is to produce synthetic data on a data set with a non-pass category so that the resulting data is balanced with data in the non-pass category. The results are shown in Figure 8. Data Balancing Process Results show that the synthetic data produced is data in the non-pass category so that the amount of data that does not pass and the amount of data that passes is balanced.

The results of the balancing process data are expected to improve the performance of the prediction model for student graduation in a course in the learning process.

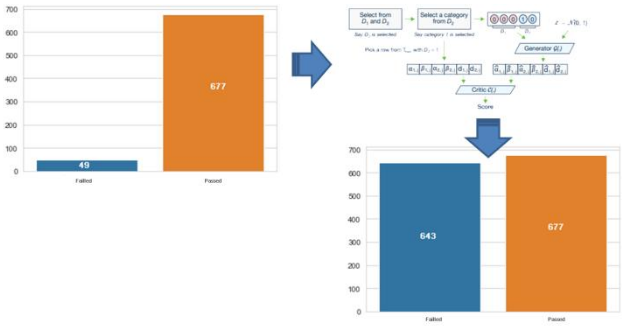

Fig. 8. Data Balancing Process Results

4.4 Result

At this stage, the researcher will present a series of prediction results carried out using a decision tree approach which is represented in the form of a confusion matrix. The results obtained will also display a performance table of the decision tree model which is measured by accuracy, precision, recall, and f1-score. The prediction results can be seen in Fig. 9. Decision Tree Prediction Result.

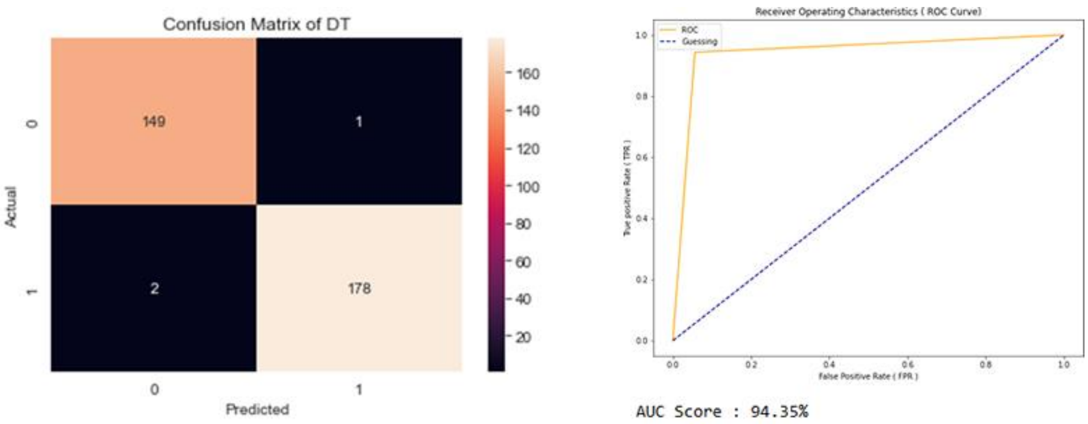

Fig. 9. Decision Tree Prediction Results

Table 1. Matrix Evaluation Decision Tree

	precision	recall	f1-score	support
Failed (0)	0.99	0.99	0.99	150
Passed (1)	0.99	0.99	0.99	180
Accuracy			0.99	330

The results are shown in Fig. 9. Decision Tree Prediction Results that the classification model using the decision approach can predict students who do not pass (Category 0/positive) very accurately (99%), whereas the classification model can also predict students who pass (Category 1/negative) very accurately accurate (99%). The results achieved by the decision tree model in predicting student graduation in a course are very significant The performance of the prediction model using the decision tree approach is shown in table 1, resulting in an accuracy value of 99%, which means that the model's ability to make predictions on data with categories passed by 99%.

4.5 Discussion

To improve student academic performance in the learning process and reduce dropout rates, a system is needed that can predict student success at the beginning of the learning evaluation process, the aim is to identify the risk of student failure in completing the learning process. The expected results from the process of predicting student success in the learning process can be used as a basis for decision making to assist students in completing the learning process that has been carried out. This research aims to test a prediction model using a decision tree approach, where the results are expected to produce very significant accuracy and precision values. Accurate prediction results will certainly help higher education providers in reducing the dropout rate caused by students who do not pass or fail in a course. However, in further research, it is necessary to carry out research using other classification models, so that the results from research using the decision tree model can be compared so that the process of generalization and justification of the prediction results can be carried out well. In the process of handling unbalanced data, it can also be done using other methods such as SMOTE, down sampling or oversampling approaches where the resulting dataset can be used to test the decision tree model so that the resulting prediction process can show differences.

4.6 Conclusion and Future Work

In this research, the use of the decision tree model in predicting course graduation in a course shows that the prediction results achieved have a very significant accuracy value. The findings in this research conclude that the Decision Tree prediction model can be used to predict student course passing in a course in the learning process. In this research too, the results achieved for the precision value were very significant with a value of 99%, meaning that the Decision Tree prediction model had a 99% ability to predict student failure. In the future, this research can be continued by predicting student success using other classification models to compare better prediction results using models for handling unbalanced data, and also the prediction results achieved can be used as basic input in recommendation models in the field of education.

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